

How to Develop Tourism Sustainably:Based on Multi-Objective Programming Model

Summary

Juneau, Alaska, has a thriving tourism industry, which has led to a rich economy, but also to the destruction of the natural environment and the dissatisfaction of local residents. In this paper, we develop a **sustainable tourism model** to analyze the factors of sustainable tourism development, put forward suggestions on the allocation of fiscal revenue, and try to expand to more regions.

First, by reviewing literature and collecting data, we identified five factors related to tourism (unemployment rate, population size, number of start-ups, tourist tax revenue, enrollment) and three target indicators (number of tourists, median personal income of residents, carbon dioxide emissions). We processed the data without dimensionality, and calculated the correlation between each factor and the target index by using **grey correlation analysis**. After removing the two non-optimization factors, the number of population and the number of start-ups, we used the **entropy weight method** to determine the weights of the other three factors on the three target indicators, which are **0.3311, 0.3251 and 0.3437** respectively.

Secondly, we perform **polynomial curve fitting** for seven components of the data. Further, according to the constraints and fitting function, we established a **multi-objective programming model**, and finally obtained the **ratios** of "the recommended value of the total annual universal cash transfer payment, the recommended value of the annual fiscal expenditure in terms of enrollment, and the recommended value of the annual tourist tax" with the corresponding data in 2014. The three ratios were **0.2161, 0.1213 and 1.3863**, respectively. According to this, we put forward the concrete proposals of fiscal expenditure and tourist tax adjustment.

Then, we carry out **sensitivity analysis** to the model. The analysis shows that the sensitivity of the model is **moderate**, and it is most sensitive to **the median personal income of residents**. The model has also been extended to over-tourism destinations and less-visited destinations respectively, which reflects certain scalability through data adjustment and changes in planning models.

In general, our model has good practicability and scalability, and can provide specific policy recommendations for Juneau City and other tourist destinations. The model provides scientific basis for fiscal budget and tax adjustment through quantitative analysis and multi-objective planning.

Keywords: Grey correlation analysis; Entropy weight method; Curve fitting; Multiobjective programming

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1 Introduction

1.1 Background

Juneau, Alaska, is a tourist destination known for its natural scenery, but in recent years, with the surge in the number of tourists, especially cruise ship tourists, the local has faced the challenge of overtourism. In 2023, Juneau hosted 1.6 million cruise ship passengers, bringing significant economic revenue to the region, but also contributing to infrastructure stress, environmental degradation, and reduced quality of life for residents. Among them, the Mendenhall Glacier has accelerated its retreat due to rising temperatures and overtourism, raising concerns among local residents about its sustainable future. How to achieve sustainable development of tourism while protecting natural resources and improving the quality of life of residents has become an urgent problem to be solved in Juneau.



Figure 1: A view of Juneau

1.2 Restatement of the Problem

Through in-depth analysis and research on the background of the problem, combined with the specific constraints given, the restate of the problem can be expressed as follows:

- (1) Various factors affecting and affected by tourism are comprehensively considered to collect relevant data. Identify the optimization factors and constraints of sustainable tourism, and establish a sustainable tourism model. According to the model, a spending plan with additional revenue is proposed so that it can promote sustainable tourism development.
- (2) A sensitivity analysis of the sustainable tourism model is conducted to study which factors have the greatest impact on it.
- (3) The model is extended to other overvisited and under-visited areas respectively, and the impact of the change of location on the sustainable tourism model is analyzed. Corresponding measures are proposed by using sustainable tourism model.

- (4) Provide the Juneau Tourism Council with a one-page memorandum outlining our projections through the sustainable Tourism model, analysis of the measures, and recommendations for effective measures.

1.3 Our Work

- First, by synthesizing the actual situation related to tourism and the specific data collected, we selected five influencing factors, as well as three target indicators of establishing sustainable tourism: economic benefits, residents' quality of life and environment. Then, we use grey correlation analysis to determine the correlation of each factor to each target index, and use entropy weight method to determine the influence weight of each factor to the three target indicators.
- Secondly, the data is curvy fitted with the aim of transforming discrete data into continuous data and finding the quantitative relationship between factors and indicators. Then we set up a multi-objective programming model according to the constraint conditions, and finally solve the optimal allocation ratio of financial input.
- Then, we carried out sensitivity analysis on the model and explored the influence of different factors on the solution results of the model. The model was then extended to overvisited destinations and less-visited destinations respectively. The different effects of different locations on the model are analyzed.
- Finally, a one-page memo addressed to the Juneau Tourism Committee was completed, detailing our prediction results and the proposed effective measures to improve the local tourism industry and achieve sustainable tourism development.

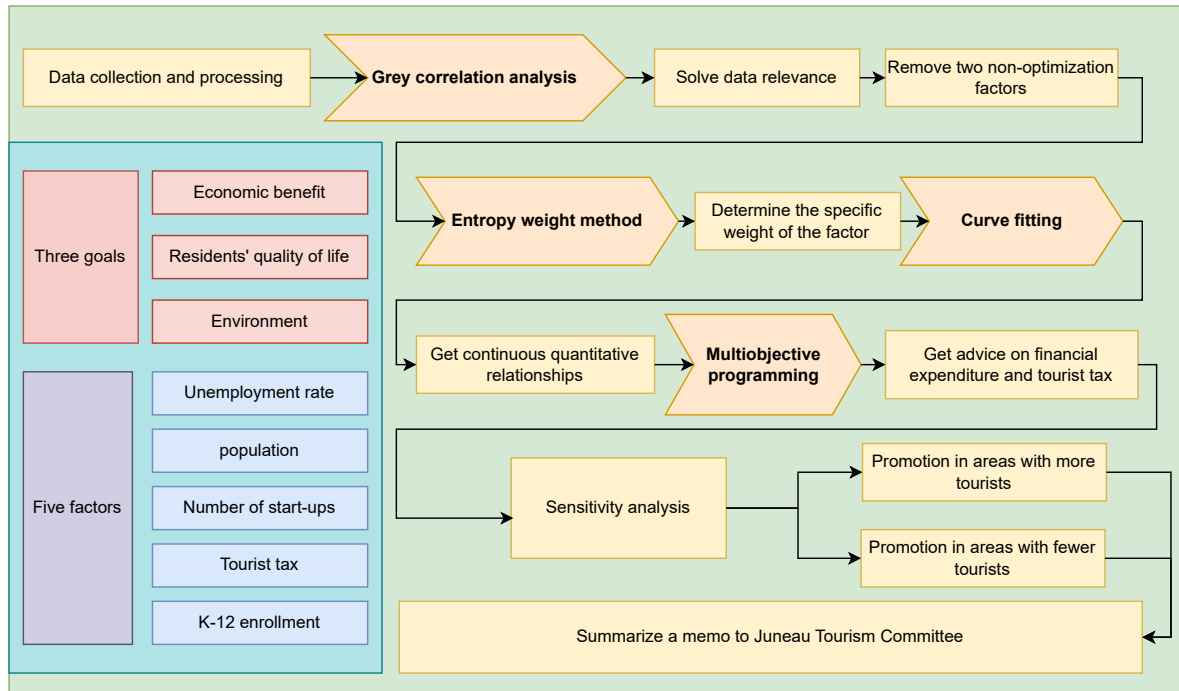


Figure 2: Flow Chart of Our Work

2 Assumptions and Justification

Considering that practical problems always contain many complex factors, first of all, we need to make reasonable assumptions to simplify the model, and each hypothesis is closely followed by its corresponding explanation:

- **Assumption 1:** The data from 2014 to 2022 can reflect the real situation of Juneau's tourism industry, and you can find patterns in it.

The data from 2014 to 2022 is selected because this period covers multiple stages of tourism development in Juneau City. By analyzing the data during this period, key factors affecting tourism development can be identified and corresponding models established to ensure the accuracy and reliability of the models.

- **Assumption 2:** The specific data index corresponding to each selected influencing factor can roughly reflect the actual impact of each influencing factor.

The specific data indicators selected by factors such as unemployment rate, population number, number of start-ups, tourist tax revenue, and enrollment number are closely related to each factor and can reflect the actual situation of each factor. At the same time, each abstract factor is reflected by a class of concrete data, which is conducive to simplifying the model and focusing on the essence of problem analysis.

- **Assumption 3: Removing the 2020 data, which was affected by the COVID-19 pandemic, makes sense for building a long-term sustainable tourism model.**

In 2020, due to the impact of the COVID-19 pandemic, there were unusual fluctuations in tourism data, which do not reflect the development of tourism in a normal year. The removal of 2020 data can avoid the interference of outliers to the model, so that the model can better reflect the long-term development trend of tourism, and provide a more reliable basis for the formulation of sustainable tourism development strategies.

- **Assumption 4: There are linear or nonlinear relationships between the target indicators and the selected factors, and these relationships can be approximated by curve fitting.**

Through curve fitting, the discrete data can be transformed into continuous functional relations, which is convenient for solving and analyzing the model and simplifying the complexity of the model. This assumption is consistent with the actual situation, because there is a certain regularity in the actual data, and it is generally not too complicated, which provides a feasible method for the construction of the model.

- **Assumption 5: It is assumed that the allocation of government expenditure has a significant impact on tourism development, and this impact can be optimized by multi-objective planning model.**

The government's financial expenditure can directly affect the construction of tourism infrastructure and the level of public services, which has an important impact on the development of all aspects of tourism. This ensures the usefulness and guidance of the model.

- **Assumption 6: It is assumed that within the time range of the model, the natural environment and policy environment of Juneau City are relatively stable, and there will be no major changes, such as natural disasters and major policy adjustments.**

This assumption ensures the stability and reliability of the model, makes the model better reflect the long-term development trend of tourism, and provides a more reliable basis for the formulation of sustainable tourism development strategies.

- **Assumption 7: It is assumed that Juneau's tourism infrastructure and public service capacity will not change significantly in the short term to meet current tourism demand.**

This assumption ensures the applicability and practicability of the model, so that the model can better reflect the current situation of the tourism industry, and provide a more reliable basis for formulating sustainable tourism development strategies.

3 Notations

Some important mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Explanation
$x_i^t(0)$	The value of the first t target indicator after the start of the i statistics
$x_i(j)$	The value of the j factor after the start of the statistics i
$x_i^1(j)$	Start to calculate the normalized value of the j factor after i year
$\zeta^1(j)$	Grey correlation coefficient between the first j factor and the number of tourists
$\zeta^2(j)$	The grey correlation coefficient between the j factor and the median personal income of residents
$\zeta^3(j)$	Grey correlation coefficient between factor j and carbon dioxide emissions
p_{ij}	The weight of the j factor
e_j	The entropy of the j factor
d_j	Redundancy of the j information entropy
ω_j	The weight of the j factor
$f_j(x)$	The number of tourists is related to unemployment rate, population size, number of start-ups, tourist tax revenue, and enrollment
$g_j(x)$	The relationship between fiscal expenditure and unemployment rate, population size, number of start-ups, tourist tax revenue, and student enrollment
$\phi_j(x)$	Relationship between carbon dioxide emissions and tourist arrivals
$\xi_j(x)$	The relationship between the median personal income of residents and the number of tourists
$\Omega_j(x)$	The relationship between the total tourism revenue and the number of tourists

4 Model Preparation

We have obtained data on a number of factors related to Juneau's tourism industry in 2014-2022, including tourist arrivals, fiscal expenditures, median personal income of residents, carbon dioxide emissions, annual average temperature, precipitation, unemployment rate, population, number of start-ups, tourist taxes, enrollment, etc. Most of the data was obtained from Juneau Economic Development Council(JEDC), Juneau website, DataCommons, Alaska Climate Research Center and other websites.

Looking back on the development and current situation of Juneau tourism, more and more attention has been paid to the sustainability of tourism development. It is believed that while pursuing the short-term economic benefits of tourism, it is also necessary to take into account the protection of the ecological environment and the guarantee of the quality of life of local residents. So we chose three target indicators: economic efficiency, quality of life, and environment. Three sets of specific data indicators were used to reflect the number of tourists, the median personal income of residents, and carbon dioxide emissions. Then, five related factors were also selected: unemployment rate, population size, number of start-ups, tourist tax revenue, and enrollment. Five specific data sets were used to reflect the average annual unemployment rate, the population of Juneau and Southeast Alaska, the number of new business applications per year, the total annual turnover of hotels and motels, and the number of students enrolled in K-12 school districts per year.

Specific data on the above target indicators and related factors can be found in the 2024 Economic Indicators Report^[1] and DataCommons^[2] published on the official website of the JEDC.

We also collected data on the total revenue generated by tourism in the Juneau region^[3].

In addition, the allocation of government financial expenditure has always been found to have a great influence on tourism development. Therefore, a model that considers the impact of fiscal spending and attempts to give guidance is considered worthy of consideration. We found data on financial allocations for education^[4]. More broadly, we look at universal cash transfers, specifically, known as the Alaska Permanent Fund Dividend(PFD) in Alaska. In a paper by Andrew Bibler, Mouhcine Guettabi, and Matthew

N. Reimer, ^{[5][6]}, a correlation between the unemployment rate and fiscal expenditure is worth noting.

Finally, we treated all 2020 data as outliers and removed them, taking into account some distortions in the data due to the impact of COVID-19 on tourism. This amendment will not be repeated later.

5 Solution to Problem 1:Sustainable tourism model

5.1 Grey correlation analysis to solve data relevance

According to the data, we take 2014 as the initial year to start the statistics, and the statistics end in 2022. (2020 is skipped).

Let $x_i^1(0)$ be the number of tourists after the statistics of i , $x_i^2(0)$ be the median personal income after the statistics of i , $x_i^3(0)$ be the carbon dioxide emission after the statistics of i , and $x_i(j)$ be the value of the first factor of j after the statistics of i . Where $1 \leq i \leq n$, $1 \leq j \leq 5$.

We use the initial value method to normalize the data:

$$x'_i(j) = \frac{x_i(j)}{x_1(j)}$$

Then, we calculate the grey correlation coefficient, denoting

$$\zeta_i^t(j) = \frac{\min_i \min_j |x_0^t(j) - x_i(j)| + \rho \cdot \max_i \max_j |x_0^t(j) - x_i(j)|}{|x_0^t(j) - x_i(j)| + \rho \cdot \max_i \max_j |x_0^t(j) - x_i(j)|} \quad (1 \leq j \leq 5, t = 1, 2, 3)$$

Where $\rho \in (0, 1)$, use this to control the accuracy of $\zeta_i^t(j)$. Experience has shown that $\rho = 0.5$ has excellent results for general problems.

Let $\rho = 0.5$.

$$\text{Denoting } \zeta^t(j) = \frac{\sum_{i=1}^m \zeta_i^t(j)}{m} \quad (t = 1, 2, 3)$$

After solving with matlab, we will sort the obtained data from large to small, and the closer to 1, the higher the correlation degree

Table 2: Ranking of grey correlation coefficient of factors and number of tourists

Factors	Number of start-ups	Tourist tax	Population	K-12 enrollment	Unemployment rate
Grey relational coefficient	0.7956	0.7895	0.6815	0.6355	0.5504

Table 3: Ranking of grey correlation coefficients of factors and median income of residents

Factors	Tourist tax	Population	K-12 enrollment	Unemployment rate	Number of start-ups
Grey relational coefficient	0.7629	0.6828	0.6724	0.6465	0.6368

Table 4: Ranking of grey correlation coefficients of factors and CO2 emissions

Factors	Unemployment rate	K-12 enrollment	Number of start-ups	Population	Tourist tax
Grey relational coefficient	0.7562	0.7258	0.7199	0.7149	0.5547

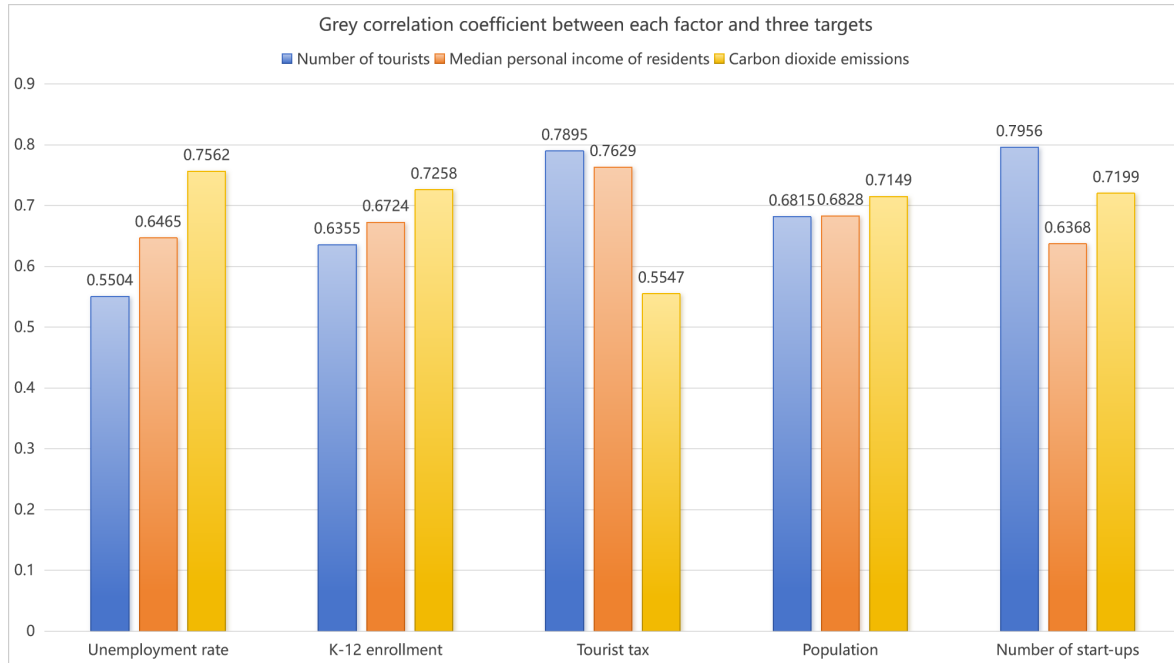


Figure 3: Grey correlation coefficient between each factor and three targets

5.2 The entropy weight method is used to determine the specific weight

We hope that through the model, we can give guidance on the fiscal expenditure in various factors. By reading the Biennial Budget Report ^[7], we noticed that the population and the number of businesses are not the main subjects of fiscal expenditure, and in fact, they are difficult to change in the short term through fiscal intervention. Therefore, we omit these two factors in the planning model and only consider the other three factors as optimization factors.

Therefore, we select the remaining three factors and comprehensively analyze the impact of these three factors on the number of tourists, median personal income of residents and carbon dioxide emissions by using the entropy weight method

Let $\zeta^1(j)$ be the grey correlation coefficient between the j th factor and the number of tourists, $\zeta^2(j)$ be the grey correlation coefficient between the j th factor and the median personal income of residents, and $\zeta^3(j)$ be the grey correlation coefficient between the j th factor and the carbon dioxide emission

The variable can be uniformly represented as $\zeta^t(j)$, where $t = 1, 2, 3$

Calculate the respective weights of the j th factor

$$p_{ij} = \frac{\zeta^t(j)}{\sum_{t=1}^3 \zeta^t(j)} \quad (t = 1, 2, 3 \quad j = 1, 2, 3)$$

Calculate the entropy of the j th factor

$$e_j = -\frac{1}{\ln n'} \sum_{i=1}^{n'} p_{ij} \ln p_{ij} \quad (t = 1, 2, 3 \quad j = 1, 2, 3)$$

Calculate the redundancy of information entropy

$$d_j = 1 - e_j \quad (j = 1, 2, 3)$$

Finally, we normalize the information entropy redundancy and get the weight of each influencing factor

$$\omega_j = \frac{d_j}{\sum_{j=1}^m d_j} \quad (j = 1, 2, 3)$$

After solving, we get the weight of each factor

Table 5: The weights of three factors		
Unemployment rate	Tourist tax	K-12 enrollment
0.3311	0.3251	0.3437

5.3 Curve fitting searches for continuous quantitative relationships

We use curve fitting to find the relationship between the data, and set $f_1(x)$, $f_2(x)$ and $f_3(x)$ as the relationship between the number of tourists and the unemployment rate, the number of students enrolled and the tourist tax revenue, respectively. $g_1(x)$ and $g_2(x)$ are the relationship between unemployment rate, enrollment and fiscal expenditure, respectively. $\phi(x)$ is the relationship between CO2 emissions and the number of tourists. $\xi(x)$ is the relationship between the median personal income of residents and the number of tourists. $\Omega(x)$ is the relationship between the total tourism revenue and the number of tourists

Among them, the relationship between unemployment rate and fiscal expenditure is proposed in a document [6], that is, $g_1(x_1) = 6.507 - 0.009 \cdot x_1$

The remaining functions are solved by curve fitting, and the fitting results are as follows:

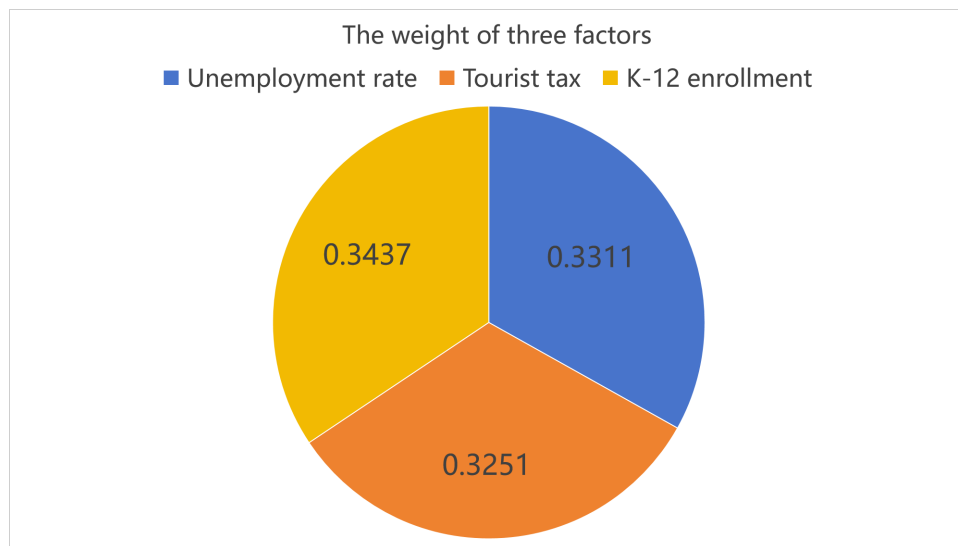


Figure 4: The weight of three factors

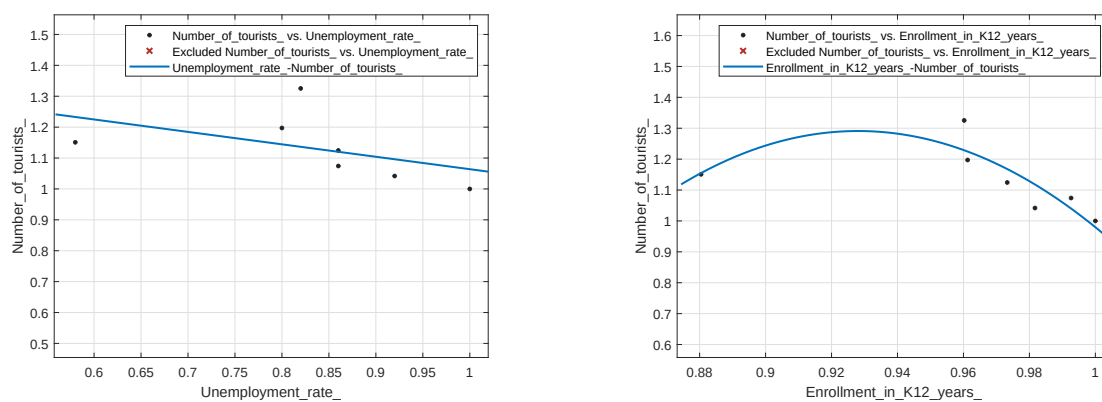


Figure 5: The fitting of unemployment rate and tourist number;The fitting of K-12 enrollment numbers with the number of tourists

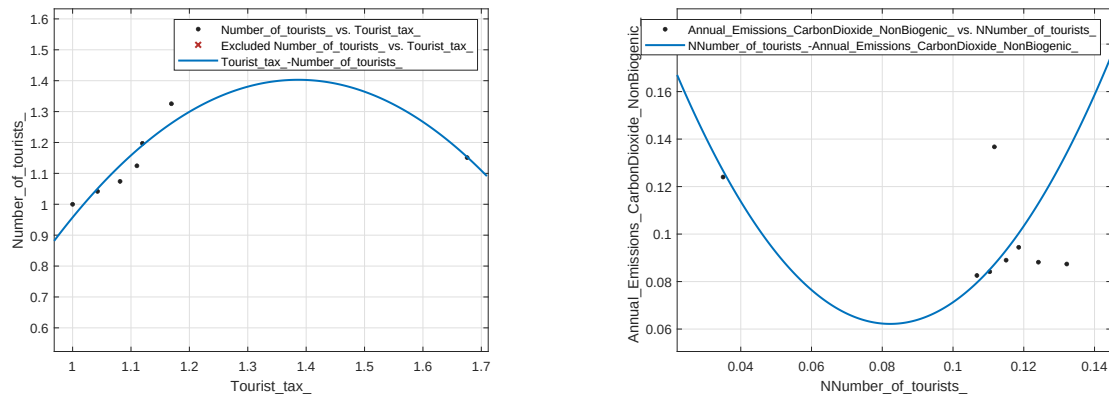


Figure 6: The fitting of tourist tax and tourist number; The fitting of tourist arrivals and CO2 emissions

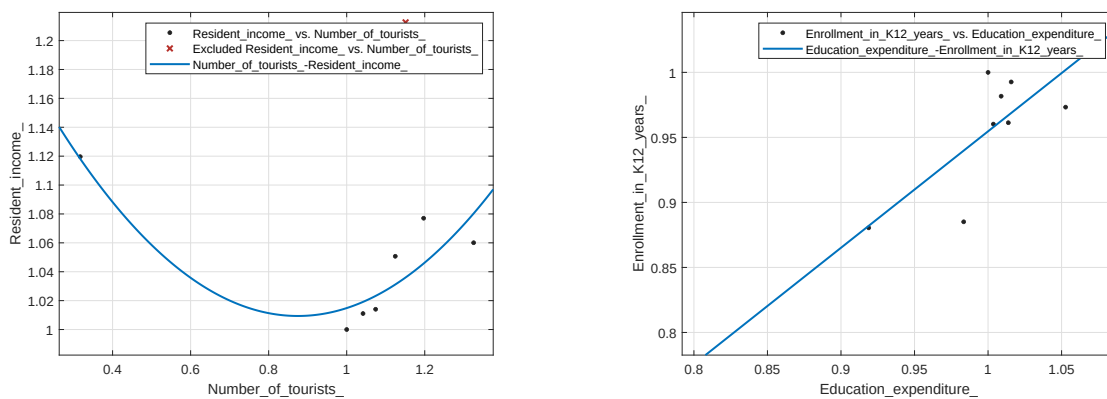


Figure 7: The fitting between the number of tourists and the personal income of residents; The fitting of education appropriations with K-12 enrollment numbers

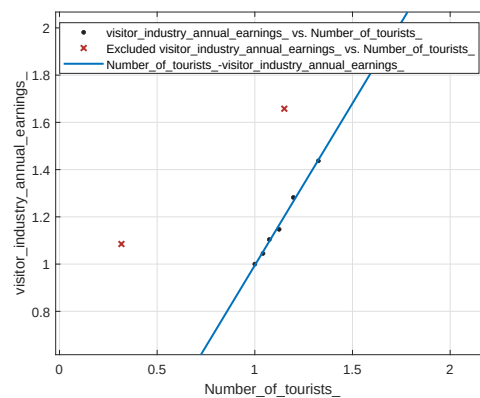


Figure 8: The fitting of the number of tourists and the total tourism revenue

Table 6: Curve fitting

	fitting curves	curve expressions	SSE	R^2	DFE	Adjusted R^2	RMSE
Unemployment rate -Number of tourists	Linear polynomial	$f_1(x) = p_1x + p_2$	0.0544	0.2332	5.0000	0.0799	0.1043
Enrollment in K12 years -Number of tourists	Quadratic polynomial	$f_2(x) = p_1 \cdot x_2 + p_2 \cdot x + p_3$	0.0194	0.7270	4.0000	0.5906	0.0696
Tourist tax -Number of tourists	Quadratic polynomial	$f_3(x) = p_1 \cdot x_2 + p_2 \cdot x + p_3$	0.0111	0.8435	4.0000	0.7653	0.0527
Number of tourists -Annual Emissions CarbonDioxide NonBiogenic	Quadratic polynomial	$\phi(x) = p_1 \cdot x_2 + p_2 \cdot x + p_3$	0.0064	0.5655	6.0000	0.4206	0.0326
Number of tourists -Resident income	Quadratic polynomial	$\xi(x) = p_1 \cdot x_2 + p_2 \cdot x + p_3$	0.0021	0.8037	4.0000	0.7056	0.0232
Education expenditure -Enrollment in K12 years	Linear polynomial	$g_2(x) = p_1x + p_2$	0.0068	0.5434	6.0000	0.4673	0.0338
Number of tourists -visitor industry annual earnings	Linear polynomial	$\Omega(x) = p_1x + p_2$	0.0008	0.9942	4.0000	0.9928	0.0139

5.4 Establish a multi-objective programming model

Since CO2 emissions are negatively targeted and tourist arrivals and median personal income of residents are positively targeted, we can build the following model:

$$maxz = \alpha \left(\sum_{j=1}^2 \omega_j f_j(g_j(x_j)) + \omega_3 f_3(x_3) \right) + \beta \xi \left(\left(\sum_{j=1}^2 \omega_j f_j(g_j(x_j)) \right) + \omega_3 f_3(x_3) \right) - \gamma \phi \left(\sum_{j=1}^2 \omega_j f_j(g_j(x_j)) + \omega_3 f_3(x_3) \right)$$

$$\begin{cases} x_j > 0 \\ \sum_{j=1}^2 x_j \leq \Omega \left(\sum_{j=1}^2 \omega_j f_j(g_j(x_j)) + \omega_3 f_3(x_3) \right) \end{cases} \quad j = 1, 2, 3$$

Where x_1 and x_2 respectively represent the total amount of universal cash transfers that the government invests in education each year in order to reduce the unemployment rate, and x_3 is the tourist tax.

By solving the multi-objective programming model, we get the results of **0.2161, 0.1213, 1.3863**. The results represent, respectively, the ratio of the recommended value of total universal cash transfers spent to reduce unemployment to the actual value of total universal cash transfers in 2014, the ratio of the recommended value of annual fiscal expenditures for enrollment to the actual value of fiscal expenditures in 2014, and the ratio of the recommended value of tourist taxes to the actual value of tourist taxes in 2014.

The above results indicate that the total amount of universal cash transfers and the annual fiscal expenditure on enrollment need to be reduced in the future, while the tourist tax needs to be appropriately increased. Specifically, the number of PFD needs to be lowered, the education allocation needs to be lowered, but the taxes on the local consumption of tourists in Juneau need to be increased, including the rental tax of hotels and motels, and the tax on cruises and sightseeing boats.

The rational explanation for this adjustment plan is that reducing the total amount of universal cash transfers can encourage residents to participate more actively in the labor market and increase employment rate, thus promoting economic diversification. At the same time, it will also help reduce the financial burden on the government and enable the government to invest more funds in other key areas such as infrastructure construction and environmental protection, thus providing better support for the sustainable development of the tourism industry. In addition, due to the diminishing marginal effect of improving the quality of education, reducing the annual fiscal expenditure on enrollment can optimize the allocation of resources, allowing the government to invest more funds

in other key areas. The proper increase of tourist tax can bring more fiscal revenue for the government, which can be used to improve infrastructure, protect the environment and improve the quality of life of residents, so as to provide better support for the sustainable development of tourism. At the same time, it can also adjust the tourism demand and avoid the negative impact of overtourism. This will help balance the relationship between tourism development and environmental protection, and achieve sustainable development of tourism.

6 Solution to Problem 2: Sensitivity analysis of the model

In order to analyze the sensitivity of the model, we change the three unknown weights in the multi-objective programming model with a certain step length, so as to obtain different financial expenditure ratio solution results.

The results obtained through matlab are shown in the figure below:

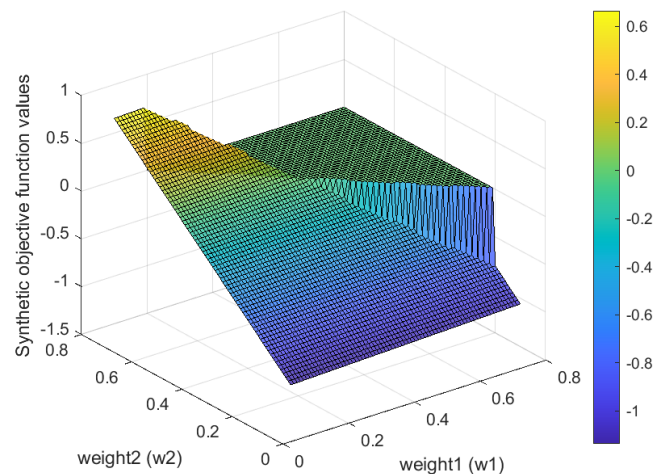


Figure 9: A graph showing the value of the synthetic objective function changing with the weight

As can be seen from the figure, weight2 has a greater impact when weight1 is small, while the impact is moderate when weight1 is large. The effect of weight1 is greater when weight2 is small, but less when weight2 is large. In general, weight2 has a greater impact on the final results, but still presents a relatively gentle linear change. Therefore, we can think that the model has moderate sensitivity, certain stability, and certain variability. The median personal income of residents represented by weight2 has the strongest influence on the model.

7 Solution to Problem 3: Promotion model for multi-visitor areas

The promotion of the sustainable tourism model in areas with more tourists is roughly the same as the prototype of the model, but the influencing factors and data must be reconsidered.

Let $x_i^1(0)$ be the number of tourists after the statistics of i , $x_i^2(0)$ be the median personal income after the statistics of i , $x_i^3(0)$ be the carbon dioxide emission after the statistics of i , and $x_i(j)$ be the value of the first factor of j after the statistics of i . Where $1 \leq i \leq n, 1 \leq j \leq m$

We use the initial value method to normalize the data:

$$x'_i(j) = \frac{x_i(j)}{x_1(j)}$$

Then, we calculate the grey correlation coefficient, denoting

$$\zeta_i^t(j) = \frac{\min_i \min_j |x_0^t(j) - x_i(j)| + \rho \cdot \max_i \max_j |x_0^t(j) - x_i(j)|}{|x_0^t(j) - x_i(j)| + \rho \cdot \max_i \max_j |x_0^t(j) - x_i(j)|} \quad (1 \leq j \leq n, t = 1, 2, 3)$$

Let $\rho = 0.5$

$$\text{Denoting } \zeta^t(j) = \frac{\sum_{i=1}^m \zeta_i^t(j)}{m} \quad (t = 1, 2, 3)$$

Then, similarly, we use the entropy weight method to calculate the effects of these m factors on the number of tourists, median personal income of residents, and carbon dioxide emissions

Let $\zeta^1(j)$ be the grey correlation coefficient between the j th factor and the number of tourists, $\zeta^2(j)$ be the grey correlation coefficient between the j th factor and the median personal income of residents, and $\zeta^3(j)$ be the grey correlation coefficient between the J TH factor and the carbon dioxide emission

Variables can be uniformly represented as $\zeta^t(j)$, $t = 1, 2, 3$

Calculate the respective weights of the j th factor

$$p_{ij} = \frac{\zeta^t(j)}{\sum_{t=1}^3 \zeta^t(j)} \quad (t = 1, 2, 3 \quad j = 1, 2, \dots, m)$$

Calculate the entropy of the j th factor

$$e_j = -\frac{1}{\ln n'} \sum_{i=1}^{n'} p_{ij} \ln p_{ij} \quad (t = 1, 2, 3 \quad j = 1, 2, \dots, m)$$

Calculate the redundancy of information entropy

$$d_j = 1 - e_j \quad (j = 1, 2, \dots, m)$$

The information entropy redundancy is normalized and the weight of each influencing factor is obtained

$$\omega_j = \frac{d_j}{\sum_{j=1}^m d_j} \quad (j = 1, 2, \dots, m)$$

After solving, we can get the weight of each factor

Then, we fit the curves of m factors and three target indicators respectively, and then substitute them into the multi-objective programming model for solving. It should be noted that since the region is in a state of overtourism, we set the three unknown weights in the objective function to 0.1, 0.45, and 0.45 respectively. In order to appropriately reduce the number of tourists in the future sustainable development planning of tourism, and comprehensively improve the economic and environmental benefits.

As a result, we can get the following constraint equation, which can be solved by plugging in specific data:

$$\begin{aligned} \max z = & 0.1 \cdot \sum_{j=1}^m \omega_j f_j(g_j(x_j)) + 0.45 \cdot \xi \left(\sum_{j=1}^m \omega_j f_j(g_j(x_j)) \right) - 0.45 \cdot \phi \left(\sum_{j=1}^m \omega_j f_j(g_j(x_j)) \right) \\ & \begin{cases} x_j > 0 \\ \sum_{j=1}^m x_j \leq \Omega \left(\sum_{j=1}^m \omega_j f_j(g_j(x_j)) \right) \end{cases} \quad j = 1, 2, \dots, m \end{aligned}$$

8 Solution to Problem 4: Promotion model for areas with few tourists

We take much the same approach to the promotion of sustainable tourism models in less visited areas. Firstly, grey correlation analysis is used to solve the correlation between local factors and the number of tourists, and then the most important factors are screened. Since the number of local tourists is small, increasing the number of tourists and maximizing the number of tourists should be regarded as the main purpose. We directly use the gray coefficient obtained in the first step to assign weights, i.e

$$\omega_j = \frac{\xi_j}{\sum_{j=1}^m \xi_j}$$

Then, we use curve fitting. Let $f_i(x)$ be the relationship between each factor and finance (proportion), and $g_i(x)$ be the relationship between the number of tourists and each factor.

The fitting equation can be obtained by solving it with matlab.

Then, according to the main purpose of increasing the number of tourists, we establish a single objective planning model:

$$\begin{aligned} \max z &= \sum_{j=1}^m g(f(x_j)) \\ \sum_{j=1}^m x_j &\leq 1 \end{aligned}$$

By plugging the specific data of the region into the model and using matlab, we can solve the optimal solution of the planning model, that is, the optimal proportion of financial investment. According to the optimal ratio, we can get the corresponding effective measures.

9 Strengths and Weaknesses

9.1 Strengths

- (1) Able to provide quantitative estimates and implementation of measures

Using grey correlation analysis and entropy weight method, the sustainable tourism model is able to provide a quantitative analysis of the considered influencing factors, including correlation and weight. With the help of curve fitting and multi-objective programming, the model gives accurate quantitative conclusions that can be implemented and provides supportive recommendations for specific fiscal budgets and taxes.

- (2) More comprehensive data analysis

The model uses a wide range of economic indicators, environmental data and demographic information to provide a more comprehensive view of Juneau's tourism industry. This integrated approach helps to understand the multifaceted impact of tourism on the region.

- (3) Robust methodological framework

The model adopts the combination of grey correlation analysis, entropy weight method and multi-objective programming. These methods are classical methods for analyzing correlation and optimization problems, which have strong applicability and robustness, and can provide satisfactory accuracy.

- (4) Adaptability and scalability

The model is designed to be adaptable to other tourist destinations, whether facing overtourism or less visited areas. Through the adjustment of relevant data, this versatility makes the model suitable for a wider range of applications and scenarios, and can make practical and effective recommendations for relevant departments.

9.2 Weaknesses

(1) Data volume limit

The model relies on historical data from 2014 to 2022 and may not fully capture the current and future dynamics of the travel industry. Less data may lead to greater deviations in grey correlation analysis and curve fitting. In addition, the removal of 2020 data due to COVID-19 may limit the ability of models to account for potential future disturbances.

(2) Incomplete consideration

In the goal index of establishing sustainable tourism, only three goals of economy, environment and people's livelihood are considered, while the in-depth examination of equity and cultural diversity is ignored. In the selection of factors, limited to the algorithm, data, model complexity and other conditions, the model initially considered only five factors. This can cause the model's estimates to deviate from reality.

(3) Sparse data points

Models use a single data point for analysis and may not capture seasonal changes or short-term fluctuations in tourism. Using a more granular approach, such as monthly or quarterly data, can provide a more detailed and accurate picture of the impact of tourism.

10 Future Work

Although the mathematical model has been established through grey correlation analysis and curve fitting, which has basically realized the factor estimation and quantitative analysis of the sustainable development of tourism, and put forward effective measures, there are still some methods and treatment methods that can be improved and are worthy of further exploration in the future. In the future, the model should be promoted in three aspects: data enhancement, algorithm optimization and empirical research.

First, the data should be better updated and expanded. The data needs to be supplemented and updated to ensure that the models reflect the latest tourism dynamics. In particular, adding data from 2023 and beyond to make up for the lack of data volume.

Moreover, additional data dimensions are needed: in addition to existing economic, environmental and demographic data, multidimensional data such as socio-cultural factors, tourist satisfaction, and tourism service quality should also be taken into account to assess the sustainability of tourism in a more comprehensive manner.

In addition, seasonal and short-term data should also be looked at, and using monthly or quarterly data for analysis is an excellent idea to capture seasonal changes and short-term fluctuations in tourism, providing a more detailed and accurate picture of the impact of tourism.

More importantly, algorithms should be optimized and enhanced. It is necessary to further optimize the grey correlation analysis and curve fitting algorithm to reduce the deviation and improve the accuracy and reliability of the model. At the same time, more

target indicators, such as equity and cultural diversity, should be added to the multi-objective planning to assess the sustainability of tourism more comprehensively.

Finally, in order to improve the feasibility and credibility of these measures, the future research can conduct empirical quantitative research on the proposed measures. This includes collecting actual data through field surveys, questionnaires, experimental studies and other methods to verify the effectiveness of the measures. In addition, relevant empirical literature can be extensively consulted and cited, and the existing research results and experience can be used as reference to provide more powerful theoretical support and data basis for the implementation of measures. Through empirical research, problems that may occur during the implementation of measures can be found and solved in time to ensure the effectiveness and scientificity of policies and measures.

References

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11 Memo

To: The Tourist Council of Juneau

From: Team 2517588

Date: Overview of The Sustainable Tourism Model

Subject: January 28, 2025

Introduction

This memo outlines our findings and recommendations for developing a sustainable tourism model in Juneau, Alaska. The model aims to balance economic benefits, environmental protection, and the quality of life for residents while promoting sustainable tourism.

Key Findings

1. Factors and Weights:

Unemployment Rate: Weight = 0.3311 ; Tourist Tax: Weight = 0.3251 ; K-12 Enrollment: Weight = 0.3437

2. Fiscal Expenditure and Tax Adjustments:

Universal Cash Transfers: Recommended reduction to 21.61% of the 2014 value.

Education Expenditure: Recommended reduction to 12.13% of the 2014 value.

Tourist Tax: Recommended increase to 138.63% of the 2014 value.

Recommendations

1. Reduce Universal Cash Transfers:

Encourage residents to participate more actively in the labor market, increasing employment rates and promoting economic diversification. Reduce the financial burden on the government, allowing for more investment in infrastructure and environmental protection.

2. Adjust Education Expenditure:

Optimize resource allocation by reducing annual fiscal expenditure on enrollment. Invest more funds in other key areas such as infrastructure and environmental protection.

3. Increase Tourist Tax:

Generate additional fiscal revenue to improve infrastructure, protect the environment, and enhance the quality of life for residents. Adjust tourism demand to mitigate the negative impacts of overtourism, balancing tourism development with environmental sustainability.

Conclusion

Implementing the recommended adjustments to fiscal expenditure and tourist taxes will promote sustainable tourism development in Juneau. These measures will help balance economic growth, environmental protection, and the quality of life for residents, ensuring a sustainable future for Juneau's tourism industry.

Appendices

matlab code

Here are simulation programmes we used in our model as follow.

Input matlab source:

```

Resident_income=[40324,40770,40890,42369,43431,42746,45148,48897];
Number_of_tourists=[1260842,1313579,1354379,1417754,1509488,1671049,...
    399839,1451233];
Unemployment_rate=[5.0,4.6,4.3,4.3,4.0,4.1,4.8,2.9];
Population_number=[33256,33445,33081,32729,32664,32544,32239,31834];
New_business_application=[239,274,270,258,300,255,303,293];
Enrollment_in_K12_years=[4751,4664,4716,4624,4567,4562,4205,4183];
Tourist_tax=[32071,33439,34677,35603,35906,37496,37829,53740];
Visitor_industry_annual_earnings=[71831,75068,79295,82398,92094,103225,...
    77938,119080];
Education_expenditure=[37808039,38144252,38402330,39801982,38328574,...
    37944127,37181939,34740707];
r=sum(Resident_income);
nu=sum(Number_of_tourists);
u=sum(Unemployment_rate);
p=sum(Population_number);
ne=sum(New_business_application);
e=sum(Enrollment_in_K12_years);
t=sum(Tourist_tax);
Resident_income_ =Resident_income/40324;
Number_of_tourists_ =Number_of_tourists/1260842;
Unemployment_rate_ =Unemployment_rate/5.0;
Population_number_ =Population_number/33256;
New_business_application_ =New_business_application/239;
Enrollment_in_K12_years_ =Enrollment_in_K12_years/4751;
Tourist_tax_ =Tourist_tax/32071;
Education_expenditure_ =Education_expenditure/37808039;
min_diffu = min(abs(Number_of_tourists_ - Unemployment_rate_));
max_diffu = max(abs(Number_of_tourists_ - Unemployment_rate_));
ucc = (min_diffu + 0.5*max_diffu) * ones(1, 8) ./ (abs( ...
    Number_of_tourists_ - Unemployment_rate_) + 0.5 * max_diffu)
min_diffp = min(abs(Number_of_tourists_ - Population_number_));
max_diffp = max(abs(Number_of_tourists_ - Population_number_));
pcc = (min_diffp + 0.5*max_diffp) * ones(1, 8) ./ (abs( ...
    Number_of_tourists_ - Population_number_) + 0.5 * max_diffp)
min_diffne = min(abs(Number_of_tourists_ - New_business_application_));
max_diffne = max(abs(Number_of_tourists_ - New_business_application_));
necc = (min_diffne + 0.5*max_diffne) * ones(1, 8) ./ (abs( ...
    Number_of_tourists_ - New_business_application_) + 0.5 * max_diffne)
min_diff e = min(abs(Number_of_tourists_ - Enrollment_in_K12_years_));
max_diff e = max(abs(Number_of_tourists_ - Enrollment_in_K12_years_));
ecc = (min_diff e + 0.5*max_diff e) * ones(1, 8) ./ (abs( ...
    number_of_tourists_ - Enrollment_in_K12_years_) + 0.5 * max_diff e)
min_diff t = min(abs(Number_of_tourists_ - Tourist_tax_));
max_diff t = max(abs(Number_of_tourists_ - Tourist_tax_));
tcc = (min_diff t + 0.5*max_diff t) * ones(1, 8) ./ (abs( ...

```

```

    Number_of_tourists_ - Tourist_tax_) + 0.5 * max_diffT)
ucc_ = sum(ucc) / 8
pcc_ = sum(pcc) / 8
necc_ = sum(necc) / 8
ecc_ = sum(ecc) / 8
tcc_ = sum(tcc) / 8
min_diffU = min(abs(Resident_income_ - Unemployment_rate_));
max_diffU = max(abs(Resident_income_ - Unemployment_rate_));
Ucc = (min_diffU + 0.5*max_diffU) * ones(1, 8) ./ (abs( ...
    Resident_income_ - Unemployment_rate_) + 0.5 * max_diffU)
min_diffP = min(abs(Resident_income_ - Population_number_));
max_diffP = max(abs(Resident_income_ - Population_number_));
Pcc = (min_diffP + 0.5*max_diffP) * ones(1, 8) ./ (abs( ...
    Resident_income_ - Population_number_) + 0.5 * max_diffP)
min_diffNE = min(abs(Resident_income_ - New_business_application_));
max_diffNE = max(abs(Resident_income_ - New_business_application_));
NEcc = (min_diffNE + 0.5*max_diffNE) * ones(1, 8) ./ (abs( ...
    Resident_income_ - New_business_application_) + 0.5 * max_diffNE)
min_diffE = min(abs(Resident_income_ - Enrollment_in_K12_years_));
max_diffE = max(abs(Resident_income_ - Enrollment_in_K12_years_));
Ecc = (min_diffE + 0.5*max_diffE) * ones(1, 8) ./ (abs( ...
    Resident_income_ - Enrollment_in_K12_years_) + 0.5 * max_diffE)
min_diffT = min(abs(Resident_income_ - Tourist_tax_));
max_diffT = max(abs(Resident_income_ - Tourist_tax_));
Tcc = (min_diffT + 0.5*max_diffT) * ones(1, 8) ./ (abs( ...
    Resident_income_ - Tourist_tax_) + 0.5 * max_diffT)
Ucc_ = sum(Ucc) / 8
Pcc_ = sum(Pcc) / 8
NEcc_ = sum(NEcc) / 8
Ecc_ = sum(Ecc) / 8
Tcc_ = sum(Tcc) / 8
Annual_Emissions_CarbonDioxide_NonBiogenic1 = ...
[36413.1, 38523.4, 40874.2, 38158.2, 37819.7, 92404.1, 53675.7];
Annual_Emissions_CarbonDioxide_NonBiogenic1 = ...
Annual_Emissions_CarbonDioxide_NonBiogenic1 / 35734.68671;
%
min_diffU_ = min(abs(Annual_Emissions_CarbonDioxide_NonBiogenic1_ - ...
    Unemployment_rate_(end)));
max_diffU_ = max(abs(Annual_Emissions_CarbonDioxide_NonBiogenic1_ - ...
    Unemployment_rate_(end)));
Ucc2 = (min_diffU_ + 0.5*max_diffU_) * ones(1, 7) ./ (abs( ...
    Annual_Emissions_CarbonDioxide_NonBiogenic1_ - ...
    Unemployment_rate_(end)) + 0.5 * max_diffU_)
min_diffP_ = min(abs(Annual_Emissions_CarbonDioxide_NonBiogenic1_ - ...
    Population_number_(end)));
max_diffP_ = max(abs(Annual_Emissions_CarbonDioxide_NonBiogenic1_ - ...
    Population_number_(end)));
Pcc2 = (min_diffP_ + 0.5*max_diffP_) * ones(1, 7) ./ (abs( ...
    Annual_Emissions_CarbonDioxide_NonBiogenic1_ - ...
    Population_number_(end)) + 0.5 * max_diffP_)
min_diffNE_ = min(abs(Annual_Emissions_CarbonDioxide_NonBiogenic1_ - ...
    New_business_application_(end)));
max_diffNE_ = max(abs(Annual_Emissions_CarbonDioxide_NonBiogenic1_ - ...
    New_business_application_(end)));

```

```

NEcc2 = (min_diffNE_ + 0.5*max_diffNE_) * ones(1, 7) ./ (abs( ...
    Annual_Emissions_CarbonDioxide_NonBiogenic1_ - ...
    New_business_application_(end)) + 0.5 * max_diffNE_)
min_diffE_ = min(abs(Annual_Emissions_CarbonDioxide_NonBiogenic1_ - ...
    Enrollment_in_K12_years_(end)));
max_diffE_ = max(abs(Annual_Emissions_CarbonDioxide_NonBiogenic1_ - ...
    Enrollment_in_K12_years_(end)));
Ecc2 = (min_diffE_ + 0.5*max_diffE_) * ones(1, 7) ./ (abs( ...
    Annual_Emissions_CarbonDioxide_NonBiogenic1_ - ...
    Enrollment_in_K12_years_(end)) + 0.5 * max_diffE_)
min_diffT_ = min(abs(Annual_Emissions_CarbonDioxide_NonBiogenic1_ - ...
    Tourist_tax_(end)));
max_diffT_ = max(abs(Annual_Emissions_CarbonDioxide_NonBiogenic1_ - ...
    Tourist_tax_(end)));
Tcc2 = (min_diffT_ + 0.5*max_diffT_) * ones(1, 7) ./ (abs( ...
    Annual_Emissions_CarbonDioxide_NonBiogenic1_ - ...
    Tourist_tax_(end)) + 0.5 * max_diffT_)
Ucc2_=sum(Ucc2)/8
Pcc2_=sum(Pcc2)/8
NEcc2_=sum(NEcc2)/8
Ecc2_=sum(Ecc2)/8
Tcc2_=sum(Tcc2)/8
resident_happiness_=Resident_income/r;
number_of_tourists_=Number_of_tourists/nu;
visitor_industry_annual_earnings_=Visitor_industry_annual_earnings/71831;
cc=[ucc_,tcc_,ecc_];
CC=[Ucc_,Tcc_,Ecc_];
Cc=[Ucc2_,Tcc2_,Ecc2_];
c=cc+CC+Cc;
cc_=cc./c;
CC_=CC./c;
Cc_=Cc./c;
entropy_value=(-log(3))*(cc_.*log(cc_)+CC_.*log(CC_)+Cc_.*log(Cc_));
redundancy=1-entropy_value;
entropy_weight=redundancy./sum(redundancy)
education_expenditure_=Education_expenditure/sum(Education_expenditure);
%
NNumber_of_tourists=[1219311,1275693,1260842,1313579,...
    1354379,1417754,1509488,1671049,399839];
Annual_Emissions_CarbonDioxide_NonBiogenic=[35734.68671,...
    59172,36413.1,38523.4,40874.2,38158.2,37819.7,92404.1,53675.7];
NNumber_of_tourists_=NNumber_of_tourists/sum(NNumber_of_tourists);
Annual_Emissions_CarbonDioxide_NonBiogenic_=...
Annual_Emissions_CarbonDioxide_NonBiogenic/...
sum(Annual_Emissions_CarbonDioxide_NonBiogenic);
function g1 = i1(x)
    g1 = 6.507 - 0.009 * x;
end

function g2 = i2(x)
    g2 = 7.1606 * x + 0.0593;
end

function f1 = o1(x)

```

```
f1 = -0.4024 * i1(x(1)) + 1.4663;
end

function f2 = o2(x)
    f2 = -60.0937 * (i2(x(2)))^2 + 111.5365 * (i2(x(2))) - 50.4631;
end

function f3 = o3(x)
    f3 = -2.9847 * x(3)^2 + 8.2756 * x(3) - 4.3334;
end

function f = o(x)
    f = 0.3311 * o1(x) + 0.3251 * o2(x) + 0.3437 * o3(x);
end

function k = k(x)
    k = 0.3497 * (o(x))^2 - 0.6118 * (o(x)) + 1.2770;
end

function l = l(x)
    l = 28.9061 * (o(x))^2 - 4.7544 * (o(x)) + 0.2577;
end

function z = z(x)
    z = o(x) + k(x) + l(x);
end

function m = m(x)
    m = x(1) + x(2) - 1.3715 * o(x) + 0.3767;
end

function [c, ceq] = mycon(x)
    c = [m(x)];
    ceq = [];
end

x0 = [0, 0, 0];
lb = [0, 0, 0];
ub = [];
W1 = 0.1:0.01:0.8;
W2 = 0.1:0.01:0.8;
n1 = length(W1);
n2 = length(W2);
F1 = zeros(n1, n2);
F2 = zeros(n1, n2);
F3 = zeros(n1, n2);
X1 = zeros(n1, n2);
X2 = zeros(n1, n2);
X3 = zeros(n1, n2);
FVAL = zeros(n1, n2);
for i = 1:n1
    for j = 1:n2
        w1 = W1(i);
        w2 = W2(j);
        w3 = 1 - w1 - w2;
```



```

    if w3 >= 0
        fun = @(x) w1 * -o1(x) - w2 * o2(x) - w3 * o3(x);
        [x, fval] = fmincon(fun, x0, [], [], [], [], lb, ub, ...
            @mycon, optimoptions('fmincon', 'Algorithm', 'sqp'));
        F1(i, j) = o1(x);
        F2(i, j) = o2(x);
        F3(i, j) = o3(x);
        X1(i, j) = x(1);
        X2(i, j) = x(2);
        X3(i, j) = x(3);
        FVAL(i, j) = fval;
    end
end
end
[minFVAL, idx] = min(FVAL(:));
[best_i, best_j] = ind2sub(size(FVAL), idx);
best_w1 = W1(best_i);
best_w2 = W2(best_j);
best_w3 = 1 - best_w1 - best_w2;
disp('Optimal weights:');
disp(['w1 = ', num2str(best_w1)]);
disp(['w2 = ', num2str(best_w2)]);
disp(['w3 = ', num2str(best_w3)]);
disp('Optimal solution:');
disp([X1(best_i, best_j), X2(best_i, best_j), X3(best_i, best_j)]);
disp('Synthetic objective function values:');
disp(minFVAL);
figure;
surf(W1, W2, FVAL, 'FaceAlpha', 0.8);
xlabel('weight1 (w1)');
ylabel('weight2 (w2)');
zlabel('Synthetic objective function values');
title(['A graph showing the value of the synthetic objective ' ...
    'function changing with the weight']);
colorbar;

```
